



Advanced Linear Algebra: Foundations to Frontiers

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1.3.2 What is a matrix norm? ¶

fit width

A matrix norm extends the notions of an absolute value and vector norm to matrices:

Definition 1.3.2.1. Matrix norm. Let $\nu : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$. Then ν is a (matrix) norm if for all $A, B \in \mathbb{C}^{m \times n}$ and all $\alpha \in \mathbb{C}$

- $A \neq 0 \Rightarrow \nu(A) > 0$ (ν is positive definite),
- $\nu(\alpha A) = |\alpha| \nu(A)$ (ν is homogeneous), and
- $\nu(A + B) \leq \nu(A) + \nu(B)$ (ν obeys the triangle inequality).

Homework 1.3.2.1. Let $\nu : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ be a matrix norm.

ALWAYS/SOMETIMES/NEVER: $\nu(0) = 0$.

▼ Hint

▼ Answer

▼ Solution

ALWAYS.

Now prove it.

Review the proof on [Homework 1.2.2.1](#).

Let $A \in \mathbb{C}^{m \times n}$. Then

$$\begin{aligned} \nu(0) &= \langle 0 \cdot A = 0 \rangle \\ \nu(0 \cdot A) &= \langle \|\cdot\|_\nu \text{ is homogeneous} \rangle \\ 0\nu(A) &= \langle \text{algebra} \rangle \\ 0 \end{aligned}$$

Remark 1.3.2.2. As we do with vector norms, we will typically use $\|\cdot\|$ instead of $\nu(\cdot)$ for a function that is a matrix norm.